

MINI PROJECT - WEEK 03

Topic – Workforce & Sales Intelligence Dashboard

Title: Smart Decisions using Pivot + XLOOKUP

Objective

Step into the role of a Business Analyst at a retail tech company. Your job is to transform raw Amazon sales data into a dynamic Excel dashboard. You'll clean the data, apply smart lookups, summarize using Pivot Tables, and present findings with filters and KPIs to support business decisions.

Scenario

The sales department at **Amazon Retail Partners** has provided a raw dump of 100 sales transactions across multiple regions, products, and customer profiles. Management wants a report that not only shows how products and regions are performing but also helps them explore customer trends and fulfillment efficiency. You're expected to build a dashboard that combines smart lookup logic with Pivot-based summaries.

Your Role

As a reporting analyst, your deliverables include:

- Cleaning the data for consistency and completeness
 - Structuring fields like dates, regions, and categories
 - Using XLOOKUP to connect relevant details for enhanced reporting
 - Creating Pivot Tables for insights into revenue, fulfillment, and cancellations
 - Designing a visual dashboard to make insights easy to access
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Tools to Use in Excel

- Pivot Table (Row, Column, Values, Filter)
- Basic functions: SUM, COUNT, AVERAGE, IF, TEXT, IFERROR

- XLOOKUP for matching across sheets or within structured data
- Date grouping, filtering, custom columns
- Conditional formatting, slicers, and charts for visual appeal

Dataset Structure: Amazon Sales (100 records)

Task Breakdown

1. Data Cleaning & Pre-processing

- Ensure date columns (Order Date, Delivery Date, Cancel Date) are in date format
- Remove extra spaces and blanks
- Standardize Region, Payment Method, and Delivery Status entries
- Identify and tag cancelled orders using Cancel Date and Delivery Status

2. Creating Pivot Tables

- Show Total Sales Amount by Region and Product Category
- Count of Orders by Delivery Status
- Average Delivery Time = Delivery Date – Order Date
- Add filter for Payment Method and Product Category

3. Applying XLOOKUP for Cross-Referencing

- Use XLOOKUP to match Customer ID to Customer Name from a separate Customer Master Sheet (you can simulate it)
- Create a new column: **Delivery Performance**
=IF([Delivery Date]-[Order Date] <= 2, "Fast", "Slow")
- Use IFERROR to avoid errors where delivery date is missing (e.g., cancellations)

4. Advanced Reporting: KPI Metrics & Calculated Fields

- Create a field: **Cancellation Rate by Region** (Cancelled Orders / Total Orders)
- Add Calculated Field in Pivot: **Effective Sales** = Sales Amount for Delivered Orders only
- Create KPI Cards for:
 - Total Revenue
 - % of Fast Deliveries
 - Top 3 Performing Regions
 - Highest-selling Product

5. Final Dashboard Design

- Use Pivot Charts and Slicers to allow filtering by Region, Payment Method, and Delivery Status
- Highlight high-performing regions or products using Conditional Formatting
- Add a drop-down to filter sales by Month (group Order Date in Pivot)
- Layout recommendations:
 - Left Panel: Filters (Slicers)
 - Center: Pivot Charts
 - Right Panel: KPIs and Summary Cards

Brainstorming Reflection

- What is the most successful product category in terms of revenue?
- Which region has the highest order cancellation rate?
- How many orders were delivered within 2 days?
- How has the use of XLOOKUP simplified cross-sheet analysis?
- What recommendations would you give to improve delivery performance?